SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	A.Veerasanthosh	**Reg. No.:**	192525187
Section / Batch:		**Date:**	10/07/2026

Code of Conduct

I **A.Veerasanthosh (192525187)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

A.Veerasanthosh

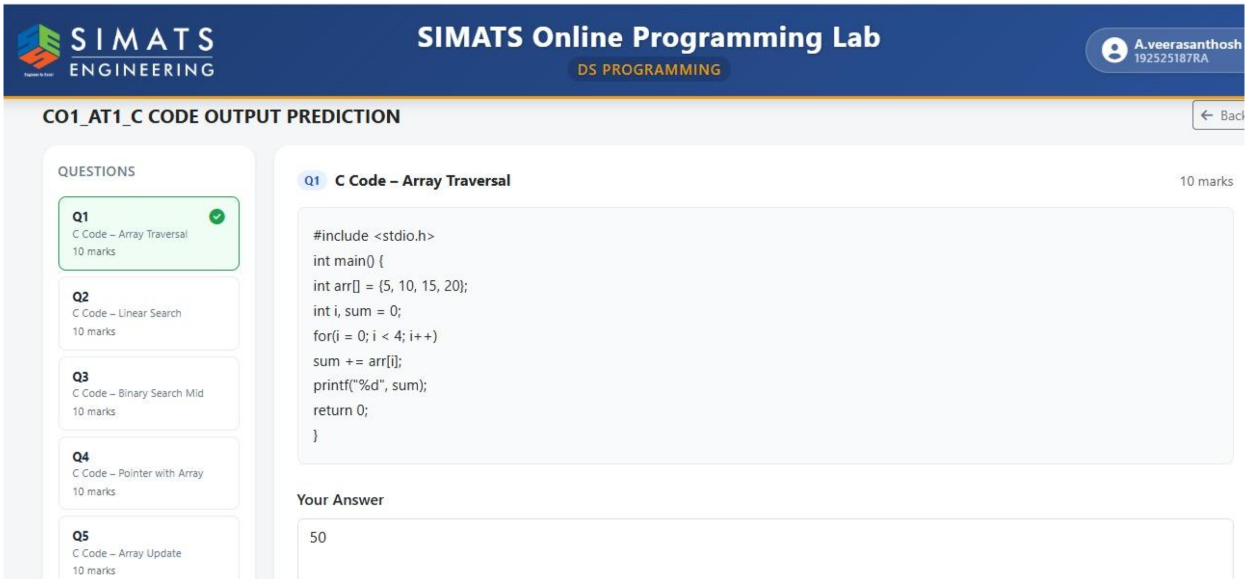
Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure / Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 - 6	20

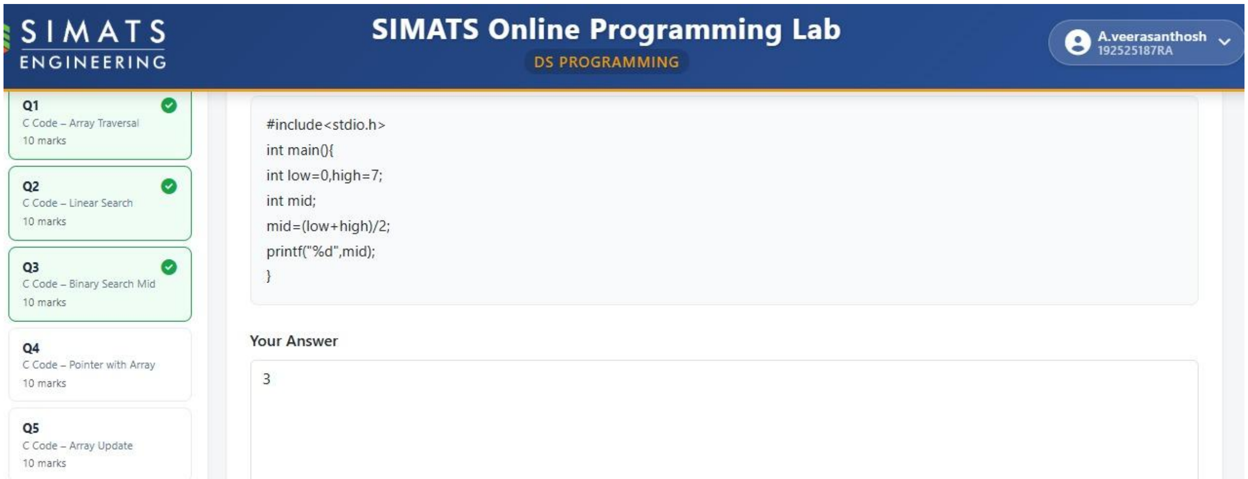
Section / Topic	Focus	Q Nos	Marks
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
		GRAND TOTAL:	100 Marks

Q1 – C Code – Array Traversal



Q2 – C Code – Linear Search

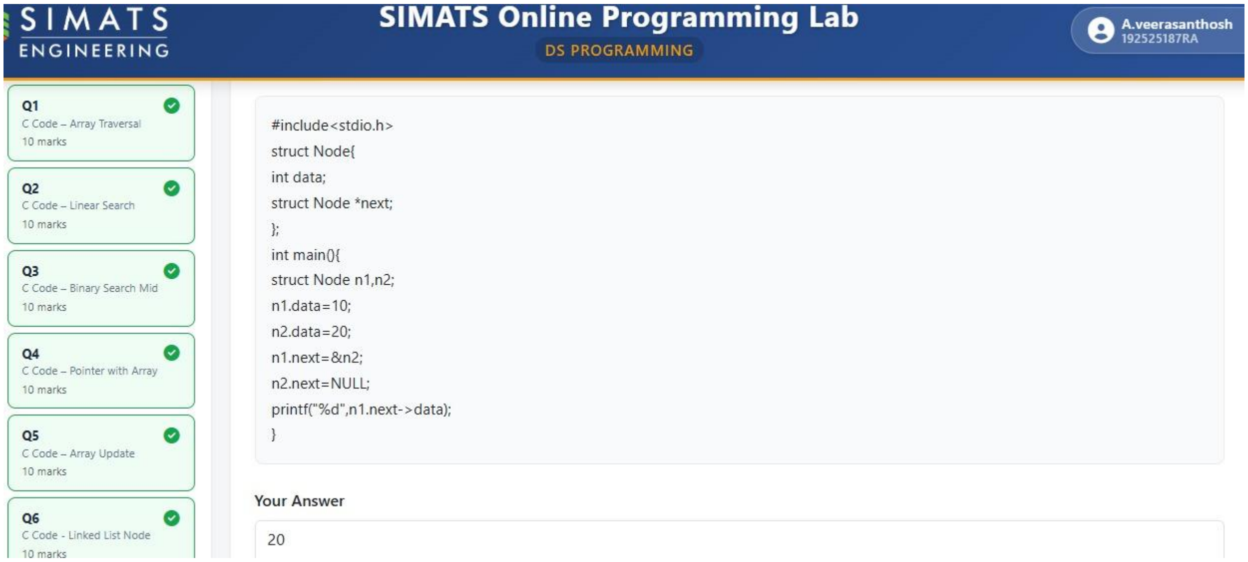
Q3 – C Code – Binary Search Mid



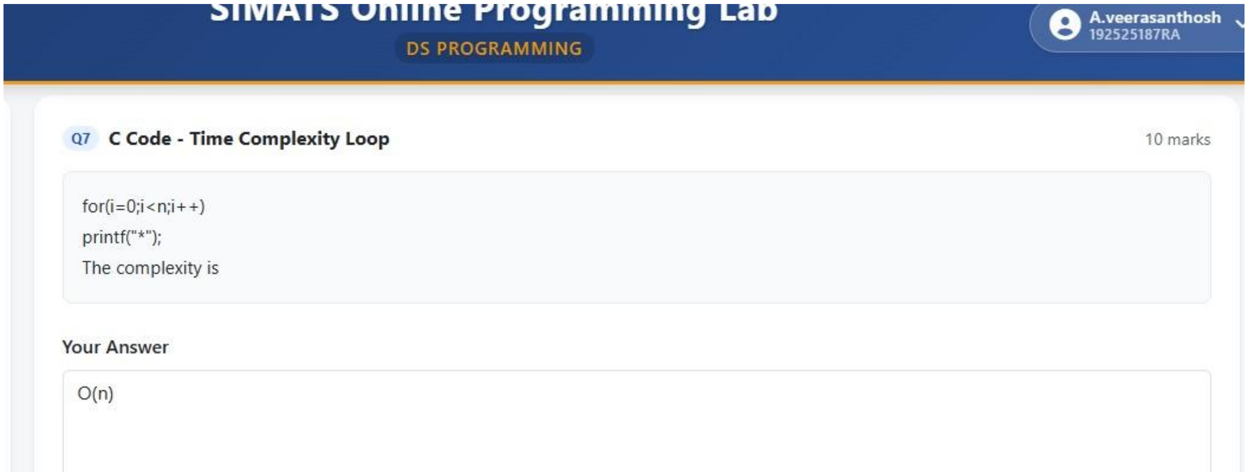
Q4 – C Code – Pointer with Array

Q5 – C Code – Array Update

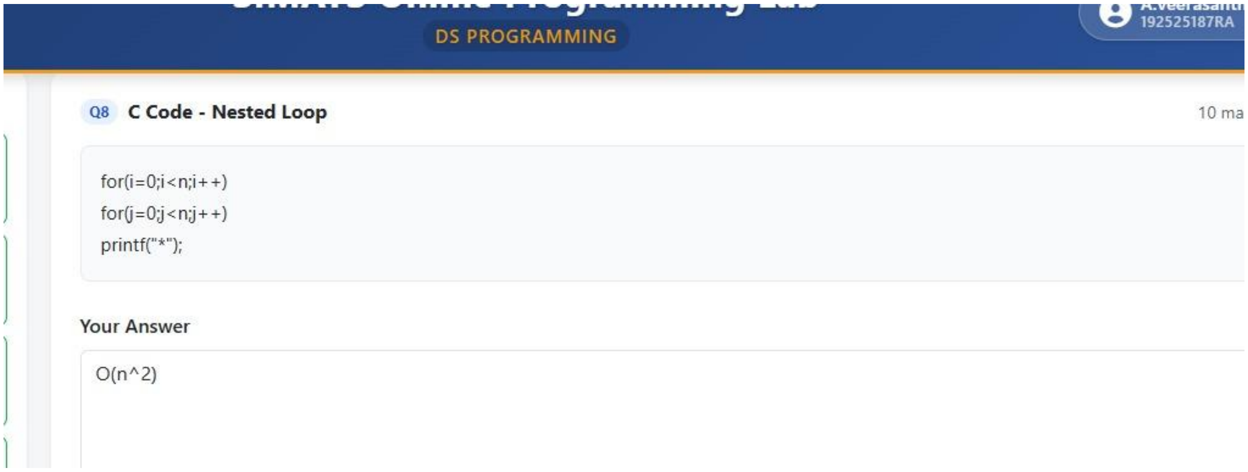
Q6 – C Code – Linked List Node



Q7 – C Code – Time Complexity Loop



Q8 – C Code – Nested Loop



Q9 – C Code – Binary Search Complexity

Q10 – C Code – Pointer Increment